AI AGENTS, MCP & APIS FOR PMS AND BAS · LESSON 5 OF 6

Claude Code Complete Tutorial: From First Prompt to Agents, MCP, Hooks and Automation

Learn Claude Code, Anthropic's agentic coding tool, from installation to advanced workflows. Work in the terminal, IDE, desktop and web; give context with CLAUDE.md; use plan mode, permissions and checkpoints; run slash commands, custom commands and Skills; add MCP servers; automate with hooks, subagents, headless mode and GitHub Actions; and follow best practices that work for PMs, analysts and engineers.

✦ 2026 | Beginner to Advanced | ⏱ 180 min

By the end of this lesson you can

- Install Claude Code and complete a task end to end in a real project
- Give Claude the right context with CLAUDE.md, files and plan mode
- Control safety with permissions, checkpoints and reviews
- Extend Claude Code with custom commands, Skills, MCP servers, hooks and subagents
- Automate work with headless mode and GitHub Actions, and know when to use each surface

What Claude Code is

Claude Code is an agent that works inside your project: it reads files, runs commands, edits code, runs tests and opens pull requests, while you watch and approve. It runs in the **terminal**, in **VS Code / JetBrains**, in the **Claude desktop app**, on the **web** (cloud sessions connected to GitHub) and on mobile, and it is used by non-engineers too: PMs build prototypes and analyses, analysts clean data and build reports.



Where you can use Claude Code.

Install and first session

STEP BY STEP **Get started**

1. Install: follow the current command on the Claude Code docs page (native installer for macOS/Linux/Windows, or `npm install -g @anthropic-ai/claude-code` with Node.js 18+).
2. Open a terminal in your project folder and run `claude`. Sign in with your Claude subscription or an API key.
3. Ask: "Explain this project's structure and how to run it." Claude reads files and summarises.
4. Ask for a small change: "Add input validation to the signup form and a test for it." Review each proposed edit and command, then approve.
5. Run `/init` to create a **CLAUDE.md** with build commands and conventions for future sessions.

Context: CLAUDE.md and how to brief Claude

`CLAUDE.md` files are loaded automatically: the project one (shared in git), personal ones (`~/.claude/CLAUDE.md`) and folder-level ones. Put in them: how to build and test, code style, architecture notes, "never do" rules, and how to publish. Keep them short and specific.

Good prompts give the goal, constraints and how to verify:

```
Goal: add a "download CSV" button to the orders page.
Constraints: reuse the existing Table component; no new dependencies; Indian date format.
Done when: the button downloads the filtered rows, a unit test covers the CSV builder, and
`npm test` passes.
Plan first, then implement.
```

Use `@path/to/file` to point at files, paste screenshots for UI bugs, and ask Claude to read docs URLs.

Plan mode, permissions and checkpoints

Feature	What it does
Plan mode (Shift+Tab to cycle modes)	Claude researches and proposes a plan without editing; you approve before it acts
Permission modes	Ask before edits/commands (default), auto-accept edits, or plan only; allowlist safe commands in `settings.json`
Checkpoints / rewind	Every edit is checkpointed; `Esc Esc` or `/rewind` restores code and conversation
`/clear`, `/compact`	Start fresh or summarise long sessions to save context
Thinking	Ask Claude to "think hard" for tricky design problems

Built-in slash commands worth knowing

Command	Use
`/init`	Create CLAUDE.md
`/model`	Switch model
`/permissions`	Manage allowed tools and commands
`/mcp`	List and authenticate MCP servers
`/agents`	Create and manage subagents
`/hooks`	Configure hooks
`/review`, `/security-review`	Review the current changes
`/cost`, `/context`	See usage and what fills the context
`/resume`	Continue a previous session

Extending Claude Code

Custom slash commands and Skills

- **Custom commands**: Markdown files in `.claude/commands/` (project) or `~/.claude/commands/` (personal); `/release-notes` can run your team's exact release-notes prompt with `$ARGUMENTS`.
- **Skills**: folders with a `SKILL.md` (instructions + optional scripts) that Claude loads automatically when relevant, e.g. "our PRD format", "brand slide style", "how to deploy". Plugins bundle commands, skills, agents, hooks and MCP servers for sharing.

MCP servers

`claude mcp add` connects tools (GitHub, Jira, Sentry, Postgres, Figma, Playwright browser, your own). Claude can then read a Figma frame and build the UI, query a read-only database to answer a data question, or open a browser to test the app.

Hooks

Hooks run your own commands at moments in Claude's workflow (before or after a tool call, when a session ends). Examples: run the formatter after every edit, block edits to `.env` files, send a Slack message when a long task finishes.

Subagents

Specialised agents with their own instructions, tools and context window, e.g. a "code reviewer", "test writer" or "data analyst". Claude delegates to them, keeping the main conversation focused.

Automation

Mode	How	Use
Headless	`claude -p "summarise yesterday's error logs" --output-format json`	Scripts, cron jobs, CI
GitHub Actions	Install the Claude GitHub app / action; mention `@claude` in an issue or PR	Implement issues, fix CI, review PRs
Cloud sessions	Start tasks on claude.ai/code against a GitHub repo	Parallel work without your laptop
Agent SDK	Python/TypeScript SDK with the same agent loop and tools	Build your own production agents

Best practices

Python

```
practices = {"Explore → plan → code → verify": "Ask Claude to read and plan before editing; finish with tests or a demo",
             "Give it a way to check its work": "Tests, a build command, screenshots, a sample input/output",
             "Small, reviewable steps": "One feature per session; commit often; use branches",
             "Keep context clean": "/clear between unrelated tasks; CLAUDE.md for durable rules",
             "Never paste secrets": "Use env vars and secret stores; review diffs before pushing"}
for k, v in practices.items(): print(f"- {k}: {v}")
```

OUTPUT

```
- Explore → plan → code → verify: Ask Claude to read and plan before editing; finish with tests or a demo
- Give it a way to check its work: Tests, a build command, screenshots, a sample input/output
- Small, reviewable steps: One feature per session; commit often; use branches
- Keep context clean: /clear between unrelated tasks; CLAUDE.md for durable rules
- Never paste secrets: Use env vars and secret stores; review diffs before pushing
```

For PMs and analysts

- **Prototype**: "Build a clickable prototype of the onboarding flow in this Figma file using our component library," then share a preview link.
- **Data analysis**: point Claude at CSVs or a read-only database and ask for cleaned data, charts and a written summary.
- **Specs and research**: read the codebase to answer "how does pricing work today?" before writing a PRD.
- **Automation scripts**: small scripts that pull data from APIs into Sheets or Slack.

Summary

KEY TAKEAWAYS

- Claude Code is an agent in your project: it reads, edits, runs and verifies, with your approval.
- CLAUDE.md, clear goals and a verification step give the best results; plan mode and checkpoints keep you safe.
- Extend with custom commands, Skills, MCP servers, hooks and subagents; automate with headless mode, GitHub Actions, cloud sessions and the Agent SDK.

Quick quiz

6 questions · tap an answer to check it

1. What is CLAUDE.md for?

A Durable project context: build/test commands, conventions and rules loaded every session

B Storing API keys

C A list of bugs

D The licence file

Answer: A. Run /init to create one.

2. Which mode lets Claude research and propose before changing anything?

A Plan mode

B Auto-accept edits

C Headless mode

D Hooks

Answer: A. Approve the plan, then let it implement.

3. How do you undo a set of Claude's edits?

- **A** Use checkpoints: Esc Esc or /rewind
- **B** Delete the repository
- **C** Reinstall Claude Code
- **D** Use /cost

Answer: A. Git commits add a second safety net.

4. What are hooks?

- **A** Your commands that run automatically at points in Claude's workflow, such as after an edit
- **B** Fishing tools
- **C** Slash commands
- **D** MCP servers

Answer: A. Use them for formatting, guards and notifications.

5. How do you run Claude Code in a script or CI job?

- **A** Headless mode with claude -p
- **B** Only through the desktop app
- **C** You cannot
- **D** Through Figma

Answer: A. Combine with --output-format json.

6. What best improves the quality of Claude's code changes?

- **A** Giving it a way to verify its work, such as tests or a build
- **B** Longer prompts with no goal
- **C** Turning off permissions
- **D** Skipping review

Answer: A. Verification closes the loop.

EXERCISE **Practice**

In a small project (or a new folder), use Claude Code to: run /init, build a small web page that reads a CSV of sales and shows a chart, add a test, create a custom command `/weekly-report` that summarises the CSV, and add one hook that runs the formatter after edits.

Show solution (worked answer)

Check: CLAUDE.md lists run/test commands; the page renders; the test passes; `.claude/commands/weekly-report.md` contains the prompt with $ARGUMENTS; the hook is in `.claude/settings.json` under PostToolUse for Edit/Write tools running your formatter.

INTERVIEW CORNER **Questions you may be asked**

Q: How do you keep an AI coding agent safe in a real codebase? Permissions and allowlists, plan mode for big changes, branches and PR reviews, tests in CI, no secrets in context, hooks that block sensitive files, and least-privilege MCP servers.

Q: How would a PM use Claude Code? Prototype features, answer "how does it work today" questions from the codebase, analyse data, write scripts that automate reporting, and review changes in plain language.

Q: Subagents vs Skills? Subagents are separate agents with their own context and tools for delegated tasks; Skills are packaged instructions and scripts that the main agent loads when a task needs them.

Q: What is headless mode used for? Running Claude Code non-interactively from scripts, scheduled jobs and CI, with structured output.

Q: How do you measure the impact of AI coding tools on a team? Lead time for changes, PR throughput and review time, defect and rollback rates, test coverage, and developer satisfaction, compared with a baseline.

Learn more

- Claude Code documentation
- Claude Code best practices (Anthropic engineering)
- Anthropic Academy courses

Practise more online

Free, trusted places to test yourself on this topic. Links open the original site in a new tab.